```
clc;
clear;
close all;
% Two Area Load Frequency Control
s = tf('s');

%% Area-1 Parameters
Tg1 = 0.08;
Tt1 = 0.30;
Tp1 = 20;
Kp1 = 120;
R1  = 2.4;

%% Area-2 Parameters
Tg2 = 0.08;
Tt2 = 0.30;
Tp2 = 20;
Kp2 = 120;
R2  = 2.4;

%% Tie-Line Synchronizing Coefficient
T12 = 0.545;

%% Governor
Gg1 = 1/(Tg1*s+1);
Gg2 = 1/(Tg2*s+1);

%% Turbine
Gt1 = 1/(Tt1*s+1);
Gt2 = 1/(Tt2*s+1);

%% Generator
Gp1 = Kp1/(Tp1*s+1);
Gp2 = Kp2/(Tp2*s+1);

%% Closed Loop of Each Area
Area1 = feedback(Gg1*Gt1*Gp1,1/R1);
Area2 = feedback(Gg2*Gt2*Gp2,1/R2);

%% Step Disturbance
t = 0:0.01:100;
DeltaPm = 0.05;
u = DeltaPm*ones(size(t));

%% Frequency Response
[df1,t] = lsim(Area1,u,t);

%% Tie-Line Effect
TieLine = tf(T12,[1 0]);
TieResponse = lsim(TieLine,df1,t);

%% Area-2 Response
df2 = lsim(Area2,TieResponse,t);

%% Plot Frequency Deviations
figure
subplot(2,1,1)
plot(t,df1,'b','LineWidth',2)
grid on
xlabel('Time (sec)')
ylabel('\Delta f_1 (Hz)')
```

```
title('Frequency Deviation of Area-1')

subplot(2,1,2)
plot(t,df2,'r','LineWidth',2)
grid on
xlabel('Time (sec)')
ylabel('\Delta f_2 (Hz)')
title('Frequency Deviation of Area-2')

%% Tie-Line Power
figure
plot(t,TieResponse,'k','LineWidth',2)
grid on
xlabel('Time (sec)')
ylabel('\Delta P_{tie} (pu)')
title('Tie-Line Power Deviation')

%% Step Characteristics
disp('Area-1 Step Characteristics')
stepinfo(df1,t)
disp('Area-2 Step Characteristics')
stepinfo(df2,t)

%% Pole-Zero Map
figure
pzmap(Area1)
grid on
title('Pole-Zero Map of Area-1')
figure
pzmap(Area2)
grid on
title('Pole-Zero Map of Area-2')

%% Root Locus
figure
rlocus(Area1)
grid on
title('Root Locus Area-1')
figure
rlocus(Area2)
grid on
title('Root Locus Area-2')

%% Bode Plot
figure
bode(Area1)
grid on
title('Bode Plot Area-1')
figure
bode(Area2)
grid on
title('Bode Plot Area-2')

%% Impulse Response
figure
impulse(Area1)
grid on
title('Impulse Response Area-1')
figure
impulse(Area2)
grid on
```

```
123  title('Impulse Response Area-2')
```